PaperFuture

Forecasting future research through trajectory-guided technical leaps

Domains: interpretable multimodal self-evolution / Agent security / AI4EDA

All model inference via APIs

A Historical preparation · Reused for each domain, subfield and cutoff T

1 Literature collection & selection

Year-stratified collection; recent two-year focus

Lexical + dense retrieval → rerank for coverage

Selected data: versioned papers + source passages

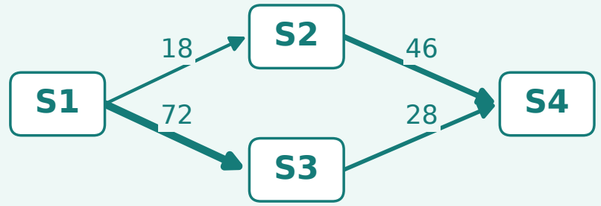
2 Weighted directed trajectory graph

Nodes: stages, papers, mechanisms

Edges: independently audited relations

Width: LLM change score (0–100)

Branches, merges, bottlenecks



Edge weight is change magnitude, not success probability

Global stage graph + round-specific source evidence

B Adaptive exploration · One candidate per round · At most four rounds in total

3 LLM leap proposal

Locate a bottleneck in the trajectory

Specify the technical change

Choose and justify leap magnitude

Idea + explicit assumptions

4 World model: simulate & discuss

Independent model forecasts
Outcomes, failures and next bottlenecks

Review and cross-model discussion
Evidence-backed objections; explicit unknowns

Reconcile and assess
Preserve novelty; assess value and feasibility

5 Adaptive decision

Good enough?
Stalled?
Round 4?

Stop

Fixed decision rules
No new candidate generation

Continue

All prior conditional experiences + retained assumptions → retrieve evidence and choose next leap

6 Idea & assessment

Main idea
Problem, mechanism, novelty

Value & feasibility
Conditions and uncertainty

Validation plan
Minimal test and sources

Best assessed candidate

Solid edge width: historical change Dashed loop: simulated feedback; never promoted to historical fact

v0.3 framework · Simulated assessments require empirical validation